STAT 423 Winter 2026 - Final Project

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Data Processing - Krish Doshi

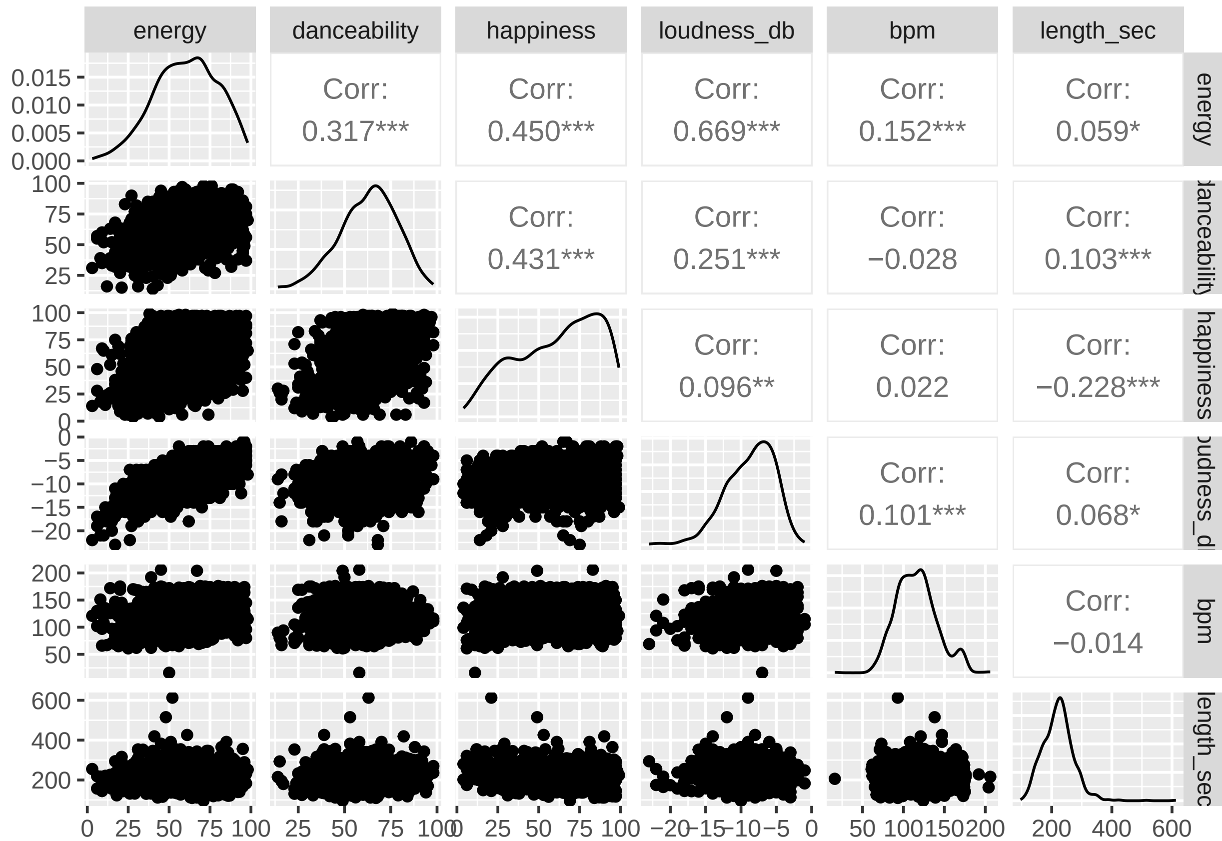
```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating = orig_data$overall_rating,
  age=orig_data$front_person_age)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified",
    orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
                      ifelse(grepl("Country", orig_data$cdr_genre), "Country",
                      ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
                      ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
                      ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other")))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy,
                danceability = orig_data$danceability, happiness = orig_data$happiness,
                loudness_db = orig_data$loudness_d_b,
                length_sec = orig_data$length_sec)


cleaned_data$year <- as.numeric(substr(orig_data$date, 1,
                    unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]
```

```
model_results <- list()


for(i in seq(1960, 2030, 10)){
  decade_label <- paste0(i - 10, "s")

  subpop <- cleaned_data[cleaned_data$year >= i - 10 &
                         cleaned_data$year < i, ]

  # --- SAFETY CHECK 1: Ensure enough rows exist ---
  # If you have < 50 rows, the interaction model will likely fail.
```

```r
  if(nrow(subpop) < 50) {
    message(paste("Skipping", decade_label, "- Not enough data points."))
    next
  }

  factor_counts <- sapply(subpop[, c("genre", "gender", "race")],
                   function(x) length(unique(na.omit(x))))

  if(any(factor_counts < 2)) {
    message(paste("Skipping", decade_label, "due to single-level factor:",
                   names(factor_counts)[factor_counts < 2]))
    next
  }

  first_model <- lm(data = subpop, rating ~ age + race + gender + genre + energy
                   + danceability + happiness + bpm + loudness_db + length_sec)

  second_model <- lm(data = subpop, rating ~ age + gender + race + age*genre
                 + age*energy + age*danceability + age*happiness + age*bpm
                 + age*loudness_db + age*length_sec + race*genre + race*energy
                 + race*danceability + race*happiness + race*bpm + race*loudness_db
                 + race*length_sec + gender*genre + gender*energy + gender*danceability
                 + gender*happiness + gender*bpm +gender*loudness_db + gender*length_sec)


  if(anova(first_model, second_model)$"Pr(>F)"[2] <= 0.1){
     optimal_model <<- step(second_model, direction="backward")
  } else {
      optimal_model <<- step(first_model, direction="backward")
  }


  model_results[[decade_label]] <- list(
    "first"   = first_model,
    "second"  = second_model,
    "optimal" = optimal_model
  )

  print(decade_label)

  summary(first_model)

  summary(second_model)

  summary(optimal_model)

}
```

```
## Skipping 1950s - Not enough data points.

## Skipping 2020s due to single-level factor: genre
```

```r
library("GGally")
ggpairs(data = cleaned_data[,c("energy", "danceability", "happiness", "loudness_db", "bpm", "length_sec
```




```r
# Loop through each decade in the model_results list
for (decade in names(model_results)) {

  # Check if the 'optimal' model exists for this decade
  if (!is.null(model_results[[decade]]$optimal)) {

    cat("\n", "========================================", "\n")
    cat("  SUMMARY FOR OPTIMAL MODEL:", decade, "\n")
    cat("========================================", "\n")

    # Print the summary
    print(summary(model_results[[decade]]$optimal))

  } else {
    cat("\nNo optimal model found for:", decade, "\n")
  }
}
```

```
##
##  ========================================
##   SUMMARY FOR OPTIMAL MODEL: 1960s
## ========================================
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.3064 -1.2960  0.1313  1.2436  4.2045
##
## Coefficients: (1 not defined because of singularities)
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  6.235892   2.001360   3.116  0.00214 **
## age                         -0.036813   0.021122  -1.743  0.08309 .
## genderNot Male              -1.820687   1.293523  -1.408  0.16101
## raceWhite                    1.604179   1.795663   0.893  0.37287
## genreJazz                    1.473039   1.332354   1.106  0.27040
## genreOther                   1.978211   0.999147   1.980  0.04926 *
## genrePop                     0.988531   0.889417   1.111  0.26788
## genreRock                    0.260014   0.629236   0.413  0.67994
## energy                      -0.005765   0.015336  -0.376  0.70743
## danceability                -0.013030   0.022465  -0.580  0.56264
## happiness                    0.004845   0.009013   0.538  0.59154
## length_sec                   0.008965   0.003791   2.365  0.01912 *
## raceWhite:genreJazz         -1.353373   2.222521  -0.609  0.54334
## raceWhite:genreOther        -3.249204   1.187123  -2.737  0.00683 **
## raceWhite:genrePop          -1.786329   0.784348  -2.277  0.02395 *
## raceWhite:genreRock                NA         NA      NA       NA
## raceWhite:energy             0.024399   0.017347   1.407  0.16132
## raceWhite:danceability      -0.046982   0.022464  -2.091  0.03791 *
## genderNot Male:danceability  0.076584   0.023552   3.252  0.00137 **
## genderNot Male:happiness    -0.030694   0.016106  -1.906  0.05830 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.797 on 178 degrees of freedom
## Multiple R-squared:  0.314,  Adjusted R-squared:  0.2446
## F-statistic: 4.527 on 18 and 178 DF,  p-value: 4.551e-08
##
##
##  ========================================
##   SUMMARY FOR OPTIMAL MODEL: 1970s
## ========================================
##
## Call:
## lm(formula = rating ~ age + race + genre + length_sec, data = subpop)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.1615 -1.0209  0.0511  1.0964  5.0519
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.250133   0.992268   5.291 2.77e-07 ***
## age         -0.033100   0.020798  -1.591 0.112830
## raceWhite   -0.430183   0.299921  -1.434 0.152800
## genreJazz   -0.547224   1.924576  -0.284 0.776402
## genreOther   0.979427   0.660001   1.484 0.139144
## genrePop    -0.607790   0.641502  -0.947 0.344375
```

```
## genreRock    0.182210   0.658709   0.277 0.782315
## length_sec   0.009494   0.002515   3.775 0.000202 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.808 on 237 degrees of freedom
## Multiple R-squared:  0.2243, Adjusted R-squared:  0.2013
## F-statistic: 9.787 on 7 and 237 DF,  p-value: 1.045e-10
##
##
##  ========================================
##   SUMMARY FOR OPTIMAL MODEL: 1980s
## ========================================
##
## Call:
## lm(formula = rating ~ energy + danceability + happiness, data = subpop)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.2377 -1.0513 -0.0178  0.9701  4.2309
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.960976   0.549218   9.033   <2e-16 ***
## energy        0.009412   0.005806   1.621   0.1064
## danceability  0.015115   0.009255   1.633   0.1038
## happiness    -0.011560   0.005495  -2.104   0.0365 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.505 on 224 degrees of freedom
## Multiple R-squared:  0.02279,    Adjusted R-squared:  0.009699
## F-statistic: 1.741 on 3 and 224 DF,  p-value: 0.1594
##
##
##  ========================================
##   SUMMARY FOR OPTIMAL MODEL: 1990s
## ========================================
##
## Call:
## lm(formula = rating ~ race + gender + energy + happiness + loudness_db,
##     data = subpop)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.2837 -0.9790 -0.0504  0.8700  4.4145
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     9.465767   0.942926  10.039  < 2e-16 ***
## raceWhite      -0.469197   0.277569  -1.690  0.09337 .
## genderNot Male  0.720941   0.253791   2.841  0.00523 **
## energy         -0.035118   0.010721  -3.276  0.00135 **
## happiness       0.016936   0.006634   2.553  0.01185 *
```

```
## loudness_db     0.311544   0.066407   4.691 6.82e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.469 on 129 degrees of freedom
## Multiple R-squared:  0.2445, Adjusted R-squared:  0.2152
## F-statistic: 8.349 on 5 and 129 DF,  p-value: 7.287e-07
##
##
##  ========================================
##   SUMMARY FOR OPTIMAL MODEL: 2000s
## ========================================
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + bpm + length_sec + age:danceability +
##     race:genre + race:danceability + race:happiness + race:bpm +
##     race:length_sec + gender:danceability + gender:happiness +
##     gender:bpm, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.80175 -0.88847 -0.00231  1.02953  2.93881
##
## Coefficients: (2 not defined because of singularities)
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 14.254799   5.689019   2.506 0.013714 *
## age                         -0.262392   0.164223  -1.598 0.113013
## genderNot Male              -0.551642   2.208098  -0.250 0.803196
## raceWhite                  -13.479191   3.724145  -3.619 0.000451 ***
## genreOther                   0.362481   1.668695   0.217 0.828443
## genrePop                     0.218854   1.608297   0.136 0.892013
## genreRock                    0.090807   1.604975   0.057 0.954986
## energy                       0.018882   0.011341   1.665 0.098826 .
## danceability                -0.083852   0.065397  -1.282 0.202519
## happiness                    0.001354   0.011393   0.119 0.905583
## bpm                         -0.014549   0.008120  -1.792 0.075985 .
## length_sec                  -0.009048   0.005130  -1.764 0.080619 .
## age:danceability             0.003737   0.002195   1.703 0.091492 .
## raceWhite:genreOther        -1.186109   0.866090  -1.369 0.173685
## raceWhite:genrePop                 NA         NA      NA       NA
## raceWhite:genreRock                NA         NA      NA       NA
## raceWhite:danceability       0.084078   0.031491   2.670 0.008759 **
## raceWhite:happiness         -0.025028   0.018899  -1.324 0.188199
## raceWhite:bpm                0.035295   0.012426   2.840 0.005385 **
## raceWhite:length_sec         0.019703   0.011520   1.710 0.090064 .
## genderNot Male:danceability -0.041970   0.024007  -1.748 0.083262 .
## genderNot Male:happiness     0.021226   0.014487   1.465 0.145771
## genderNot Male:bpm           0.019019   0.010619   1.791 0.076076 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.39 on 108 degrees of freedom
## Multiple R-squared:  0.3417, Adjusted R-squared:  0.2198
```

```
## F-statistic: 2.803 on 20 and 108 DF,  p-value: 0.0003367
##
##
##  ========================================
##   SUMMARY FOR OPTIMAL MODEL: 2010s
## ========================================
##
## Call:
## lm(formula = rating ~ gender + energy + happiness + length_sec,
##     data = subpop)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.2684 -1.0215  0.1378  0.9163  3.4440
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     4.788777   1.188134   4.031 0.000103 ***
## genderNot Male  0.459667   0.298784   1.538 0.126808
## energy         -0.024064   0.009966  -2.415 0.017405 *
## happiness       0.012193   0.007833   1.557 0.122440
## length_sec      0.009668   0.004602   2.101 0.037952 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.594 on 110 degrees of freedom
## Multiple R-squared:  0.09648,    Adjusted R-squared:  0.06363
## F-statistic: 2.937 on 4 and 110 DF,  p-value: 0.02378
```

Preliminary Exploration of Data through Visual Analysis

Plots of Individual Relationships

The plots show the relationship between the predictors and the response variable, rating.

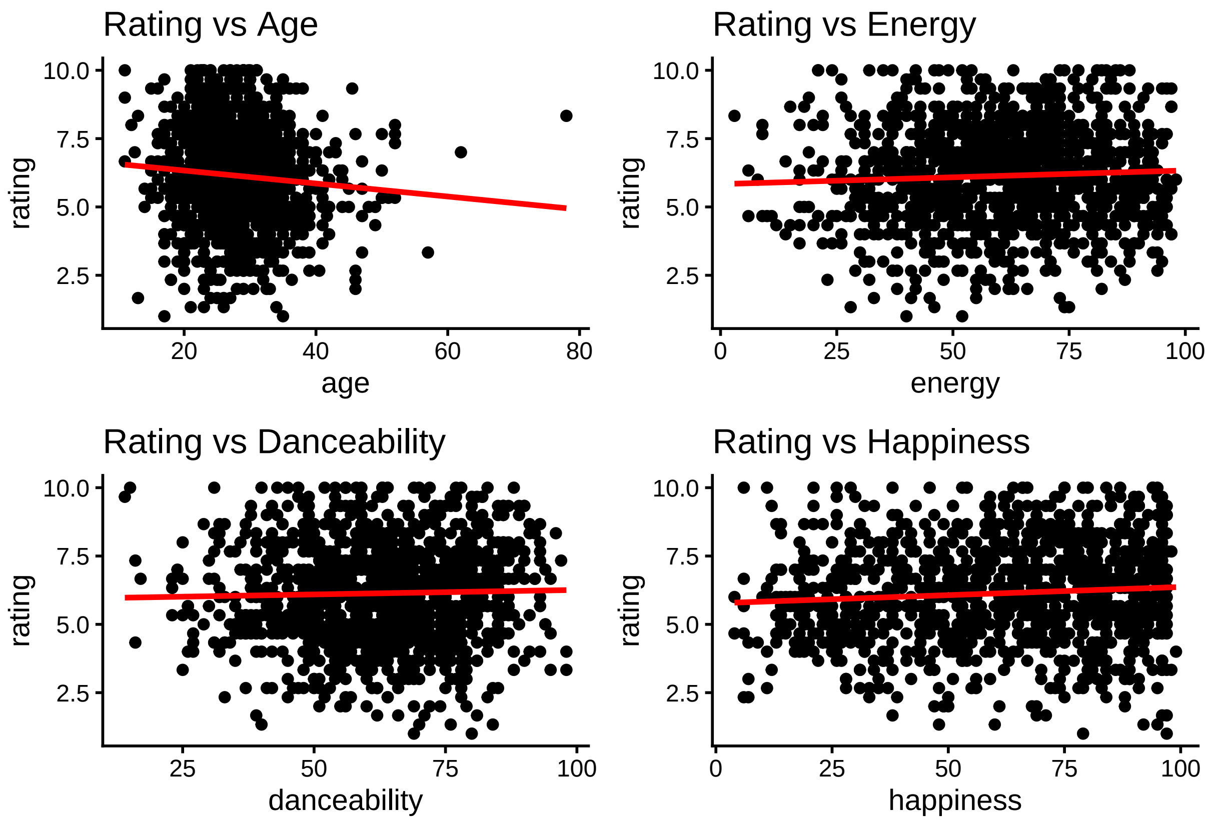
```r
library(ggplot2)
library(patchwork)
```

```
##
## Attaching package: 'patchwork'

## The following object is masked from 'package:MASS':
##
##     area
```

```r
plot_age <- ggplot(cleaned_data, aes(x = age, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Age")

plot_energy <- ggplot(cleaned_data, aes(x = energy, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Energy")
```

```r
plot_danceability <- ggplot(cleaned_data, aes(x = danceability, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Danceability")

plot_happiness <- ggplot(cleaned_data, aes(x = happiness, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Happiness")

plot_age + plot_energy + plot_danceability + plot_happiness
```

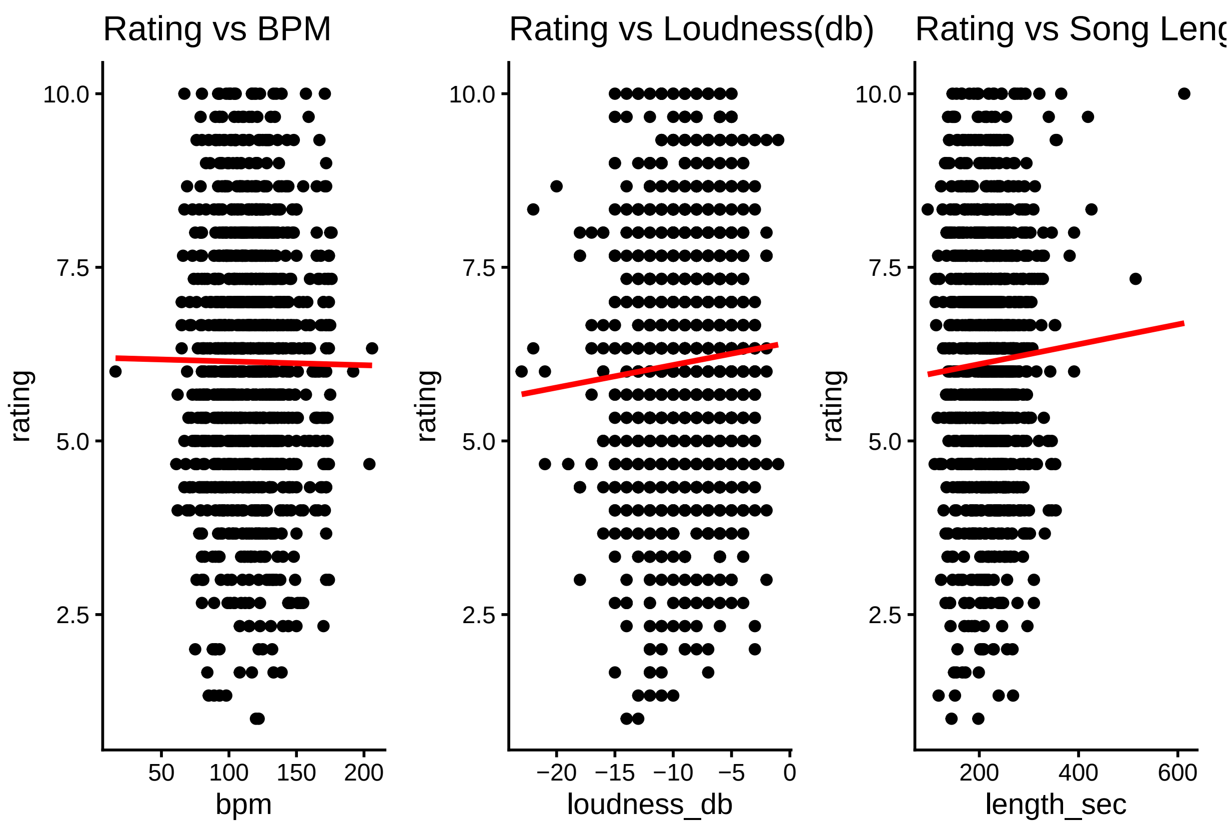
```
## `geom_smooth()` using formula = 'y ~ x'

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```




```r
plot_bpm <- ggplot(cleaned_data, aes(x = bpm, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs BPM")

plot_loudness_db <- ggplot(cleaned_data, aes(x = loudness_db, y = rating)) +
  geom_point(color = "black", size = 1.5) +
```

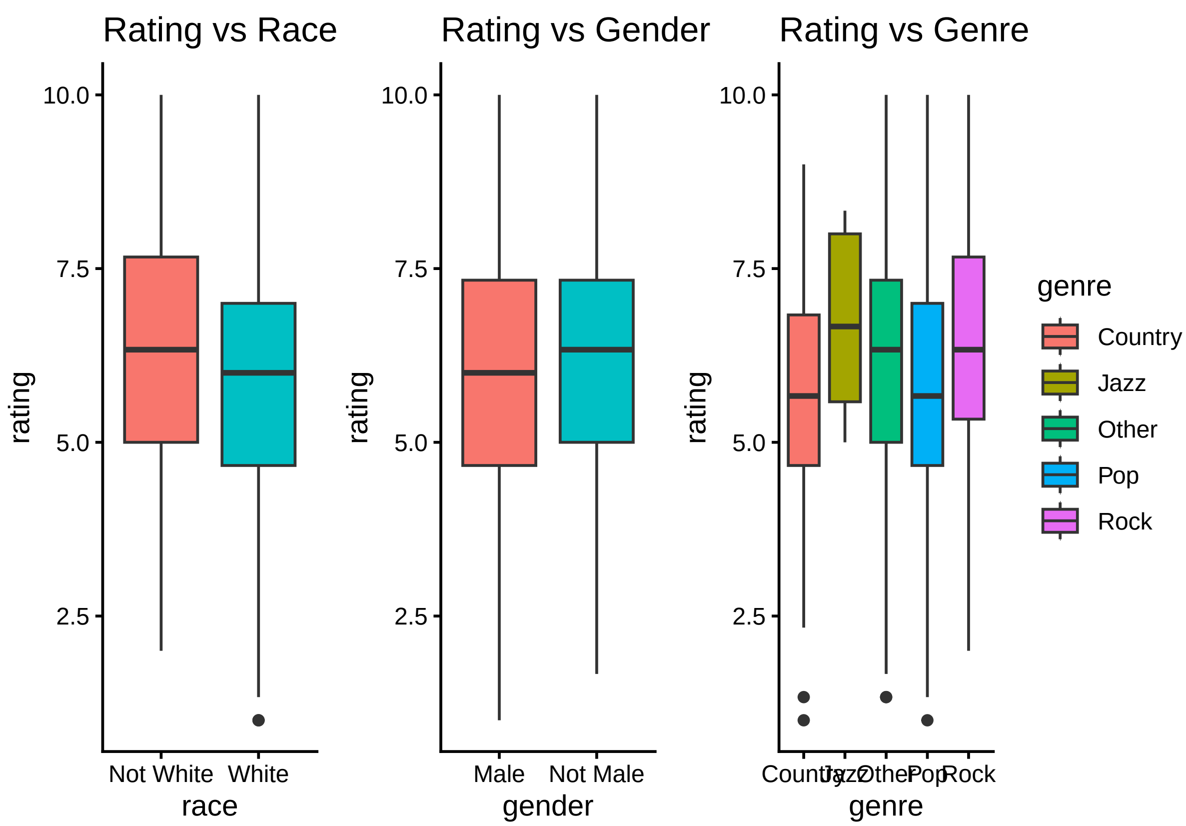
```r
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Loudness(db)")

plot_length_sec <- ggplot(cleaned_data, aes(x = length_sec, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Song Length(sec)")

plot_bpm + plot_loudness_db + plot_length_sec
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



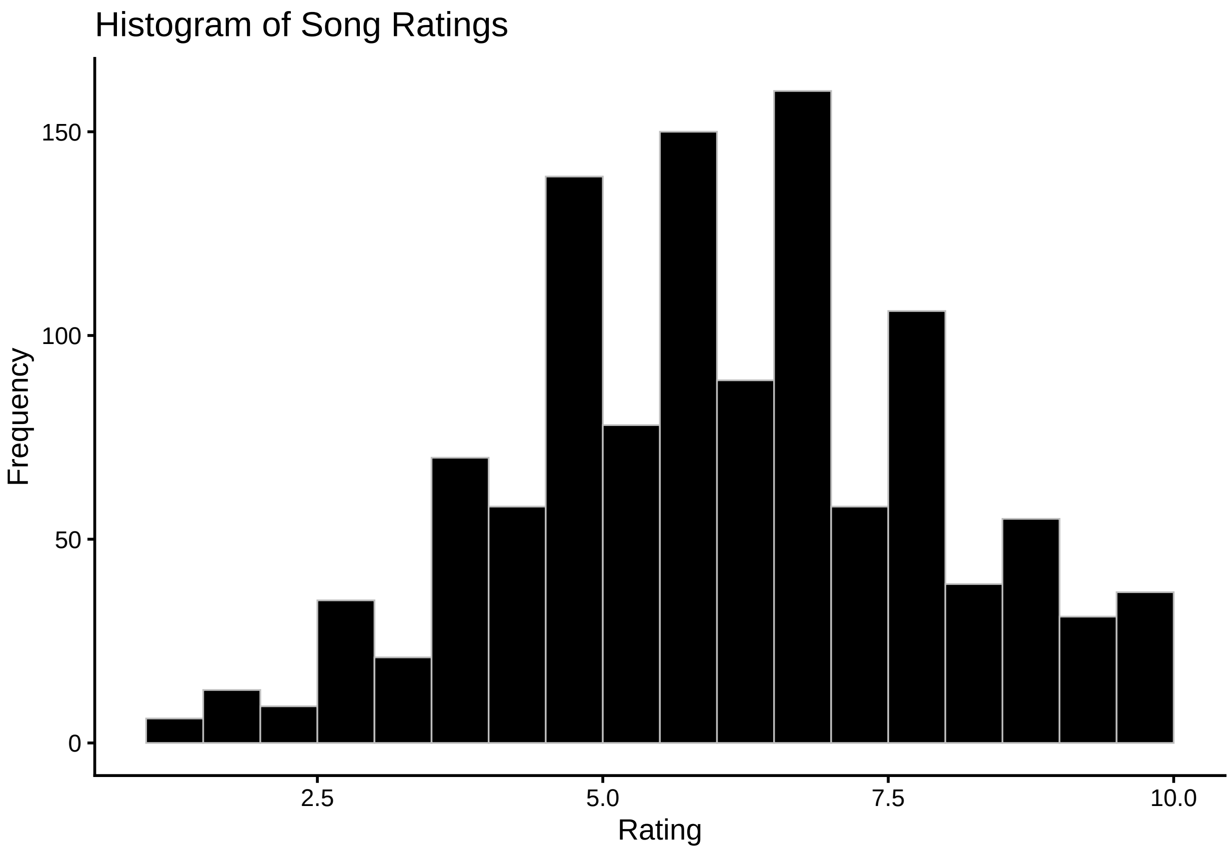

```r
plot_race <- ggplot(cleaned_data, aes(x = race, y = rating, fill = race)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Race") +
  guides(fill = "none")

plot_gender <- ggplot(cleaned_data, aes(x = gender, y = rating, fill = gender)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Gender") +
  guides(fill = "none")
```

```r
plot_genre <- ggplot(cleaned_data, aes(x = genre, y = rating, fill = genre)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Genre")

plot_race + plot_gender + plot_genre
```



As expected, banding is observed in the scatterplots because the response variable, rating, is discrete. Since race, gender, and genre are categorical predictors, the relationship with rating is displayed by boxplots instead.

```r
ggplot(cleaned_data, aes(x = rating)) +
  geom_histogram(
    binwidth = 0.5,
    boundary = 0.5,
    fill = "black",
    color = "grey",
    linewidth = 0.3
  ) +
  labs(
    title = "Histogram of Song Ratings",
    x = "Rating",
    y = "Frequency"
  ) +
  theme_classic()
```



However, the histogram of ratings appears approximately symmetric and unimodal, resembling a normal distribution. While the ratings are discrete (integer-valued) and there are many categorical predictors, the distribution spans multiple levels, creating a shape that approximates continuity. This supports the treating of rating as a continuous response variable in the regression analysis.

Plots of Grouped Relationships

Data visualizations for grouped relationships help us examine whether relationships between predictors and ratings differ across categories. These plot justify the inclusion of categorical predictors in the regression models and help identify potential interaction effects.

```
model_60s <- model_results[["1960s"]]$optimal
model_70s <- model_results[["1970s"]]$optimal
model_80s <- model_results[["1980s"]]$optimal
model_90s <- model_results[["1990s"]]$optimal
model_2000s <- model_results[["2000s"]]$optimal
model_2010s <- model_results[["2010s"]]$optimal
```

```
summary(model_60s)$r.squared
```

```
## [1] 0.3140162
```

```
summary(model_60s)$adj.r.squared
```

```
## [1] 0.244647
```

```
summary(model_70s)$r.squared
```

```
## [1] 0.2242527
```

```r
summary(model_70s)$adj.r.squared
```

```
## [1] 0.2013403
```

```r
summary(model_80s)$r.squared
```

```
## [1] 0.02278648
```

```r
summary(model_80s)$adj.r.squared
```

```
## [1] 0.009698796
```

```r
summary(model_90s)$r.squared
```

```
## [1] 0.2444873
```

```r
summary(model_90s)$adj.r.squared
```

```
## [1] 0.2152039
```

```r
summary(model_2000s)$r.squared
```

```
## [1] 0.3416655
```

```r
summary(model_2000s)$adj.r.squared
```

```
## [1] 0.2197517
```

```r
summary(model_2010s)$r.squared
```

```
## [1] 0.09648348
```

```r
summary(model_2010s)$adj.r.squared
```

```
## [1] 0.06362834
```

```r
vars_by_decade <- lapply(model_results, function(x) {
  attr(terms(x$optimal), "term.labels")
})

all_vars <- unique(unlist(vars_by_decade))

presence_matrix <- sapply(vars_by_decade, function(vars) {
  all_vars %in% vars
})

rownames(presence_matrix) <- all_vars
presence_matrix
```

```
##                       1960s 1970s 1980s 1990s 2000s 2010s
## age                    TRUE  TRUE FALSE FALSE  TRUE FALSE
## gender                 TRUE FALSE FALSE  TRUE  TRUE  TRUE
## race                   TRUE  TRUE FALSE  TRUE  TRUE FALSE
## genre                  TRUE  TRUE FALSE FALSE  TRUE FALSE
## energy                 TRUE FALSE  TRUE  TRUE  TRUE  TRUE
## danceability           TRUE FALSE  TRUE FALSE  TRUE FALSE
## happiness              TRUE FALSE  TRUE  TRUE  TRUE  TRUE
## length_sec             TRUE  TRUE FALSE FALSE  TRUE  TRUE
## race:genre             TRUE FALSE FALSE FALSE  TRUE FALSE
## race:energy            TRUE FALSE FALSE FALSE FALSE FALSE
```

```
## race:danceability     TRUE FALSE FALSE FALSE  TRUE FALSE
## gender:danceability   TRUE FALSE FALSE FALSE  TRUE FALSE
## gender:happiness      TRUE FALSE FALSE FALSE  TRUE FALSE
## loudness_db          FALSE FALSE FALSE  TRUE FALSE FALSE
## bpm                  FALSE FALSE FALSE FALSE  TRUE FALSE
## age:danceability     FALSE FALSE FALSE FALSE  TRUE FALSE
## race:happiness       FALSE FALSE FALSE FALSE  TRUE FALSE
## race:bpm             FALSE FALSE FALSE FALSE  TRUE FALSE
## race:length_sec      FALSE FALSE FALSE FALSE  TRUE FALSE
## gender:bpm           FALSE FALSE FALSE FALSE  TRUE FALSE
```

```r
rownames(presence_matrix)[rowSums(presence_matrix) == ncol(presence_matrix)]
```

```
## character(0)
```

```r
presence_df <- as.data.frame(as.table(presence_matrix))

ggplot(presence_df, aes(x = Var2, y = Var1, fill = Freq)) +
  geom_tile() +
  scale_fill_manual(values = c("black","#23DC6D")) +
  labs(x = "Decade", y = "Variable", fill = "Included") +
  theme_minimal()
```


```r
true_counts <- rowSums(presence_matrix)
true_counts
```

```
##                 age              gender                race               genre
##                   3                   4                   4                   3
##              energy        danceability           happiness          length_sec
```

```
##                   5                    3                    5                    4
##          race:genre          race:energy    race:danceability gender:danceability
##                   2                    1                    2                    2
##     gender:happiness          loudness_db                  bpm     age:danceability
##                   2                    1                    1                    1
##      race:happiness             race:bpm      race:length_sec           gender:bpm
##                   1                    1                    1                    1
```

```r
rownames(presence_matrix)[true_counts == max(true_counts)]
```

```
## [1] "energy"    "happiness"
```

```r
false_counts <- rowSums(!presence_matrix)
false_counts
```

```
##                 age               gender                 race                genre
##                   3                    2                    2                    3
##              energy         danceability            happiness           length_sec
##                   1                    3                    1                    2
##          race:genre          race:energy    race:danceability gender:danceability
##                   4                    5                    4                    4
##     gender:happiness          loudness_db                  bpm     age:danceability
##                   4                    5                    5                    5
##      race:happiness             race:bpm      race:length_sec           gender:bpm
##                   5                    5                    5                    5
```

```r
rownames(presence_matrix)[false_counts == max(false_counts)]
```

```
## [1] "race:energy"      "loudness_db"      "bpm"              "age:danceability"
## [5] "race:happiness"   "race:bpm"         "race:length_sec"  "gender:bpm"
```

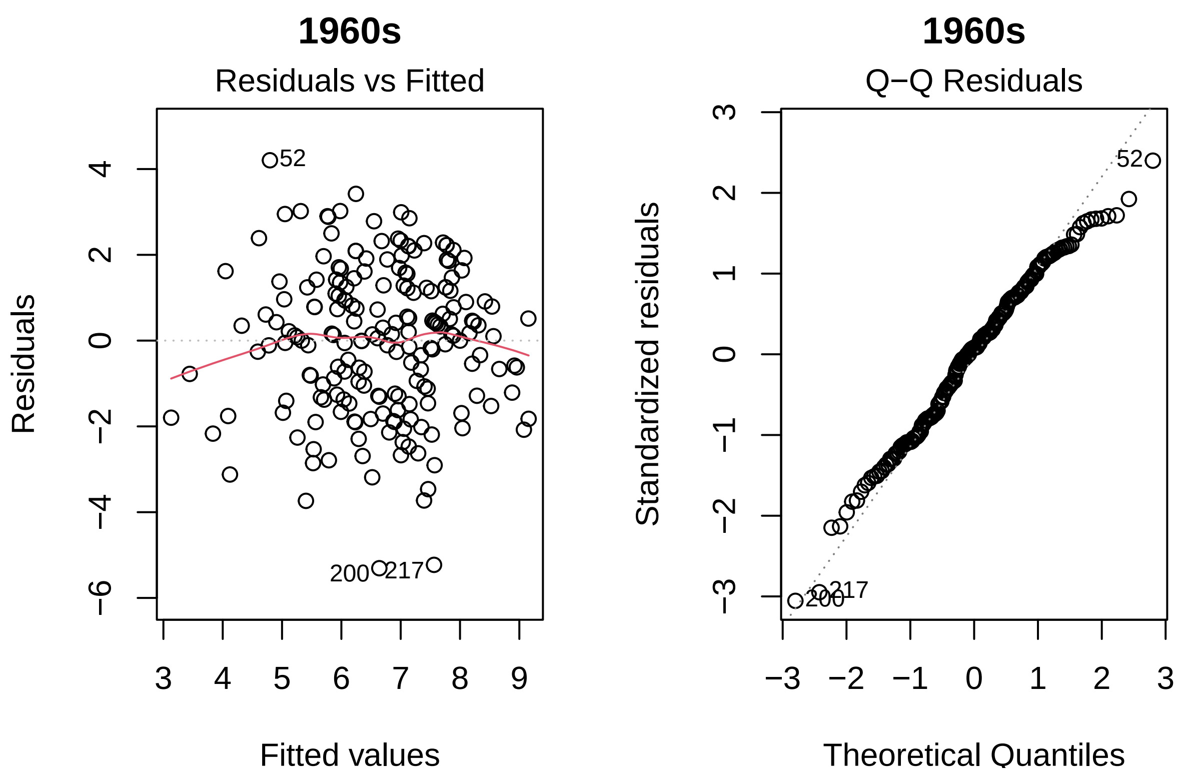
```r
summary_table <- data.frame(
  TRUE_count = rowSums(presence_matrix),
  FALSE_count = rowSums(!presence_matrix)
)

summary_table[order(-summary_table$TRUE_count), ]
```

```
##                     TRUE_count FALSE_count
## energy                       5           1
## happiness                    5           1
## gender                       4           2
## race                         4           2
## length_sec                   4           2
## age                          3           3
## genre                        3           3
## danceability                 3           3
## race:genre                   2           4
## race:danceability            2           4
## gender:danceability          2           4
## gender:happiness             2           4
## race:energy                  1           5
## loudness_db                  1           5
## bpm                          1           5
## age:danceability             1           5
## race:happiness               1           5
## race:bpm                     1           5
```

```
## race:length_sec                1           5
## gender:bpm                     1           5
```

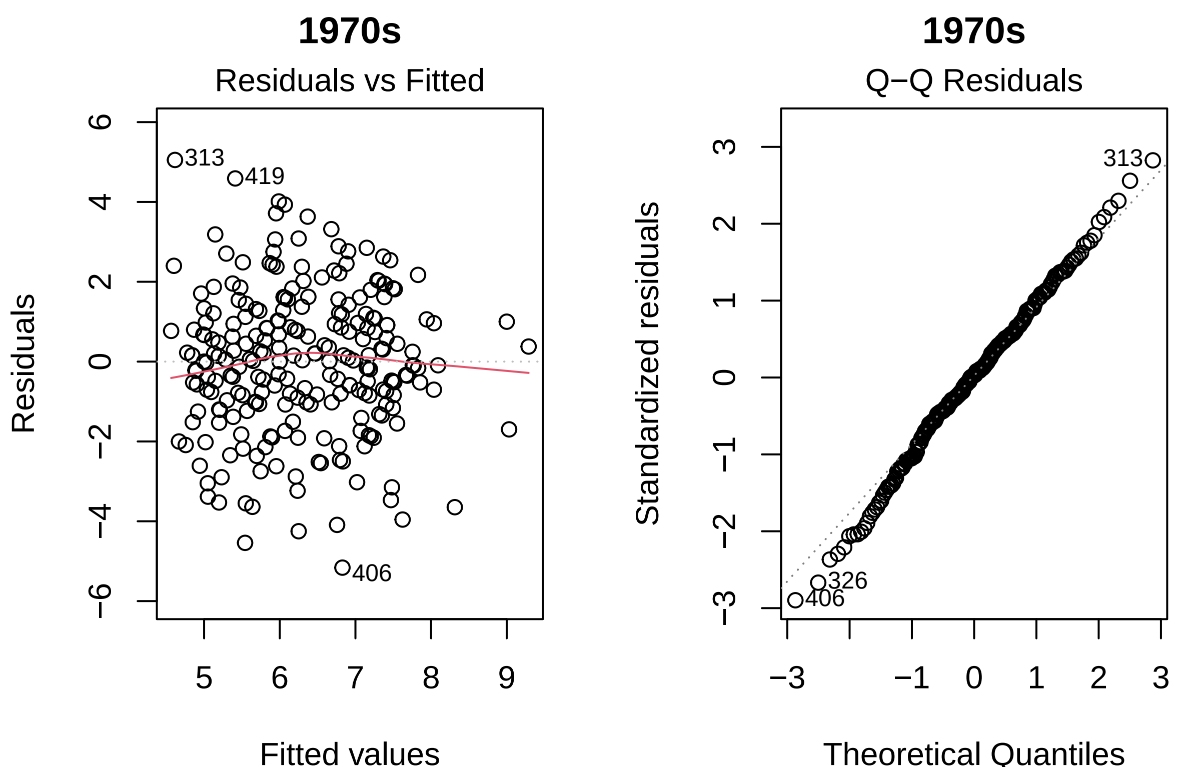
```r
par(mfrow = c(1,2))
plot(model_60s, which = 1:2, main = "1960s")
```

```
## Warning: not plotting observations with leverage one:
##   48
```

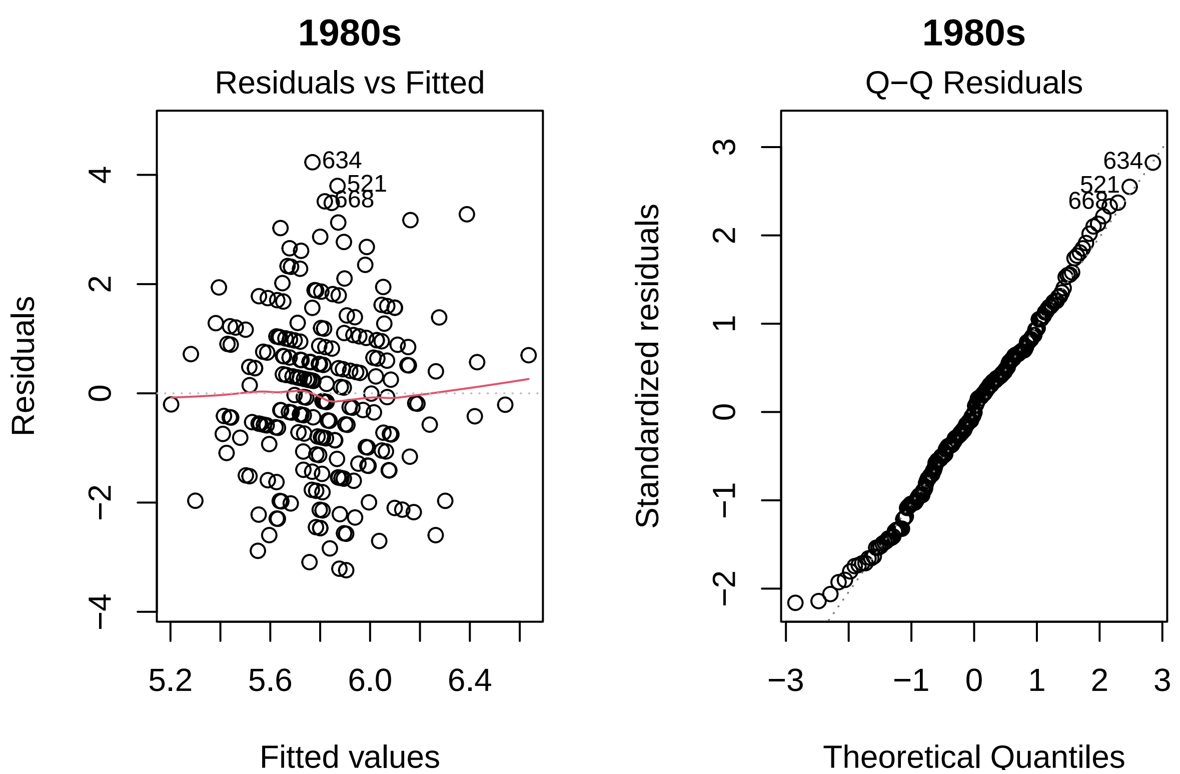



```r
plot(model_70s, which = 1:2, main = "1970s")
```

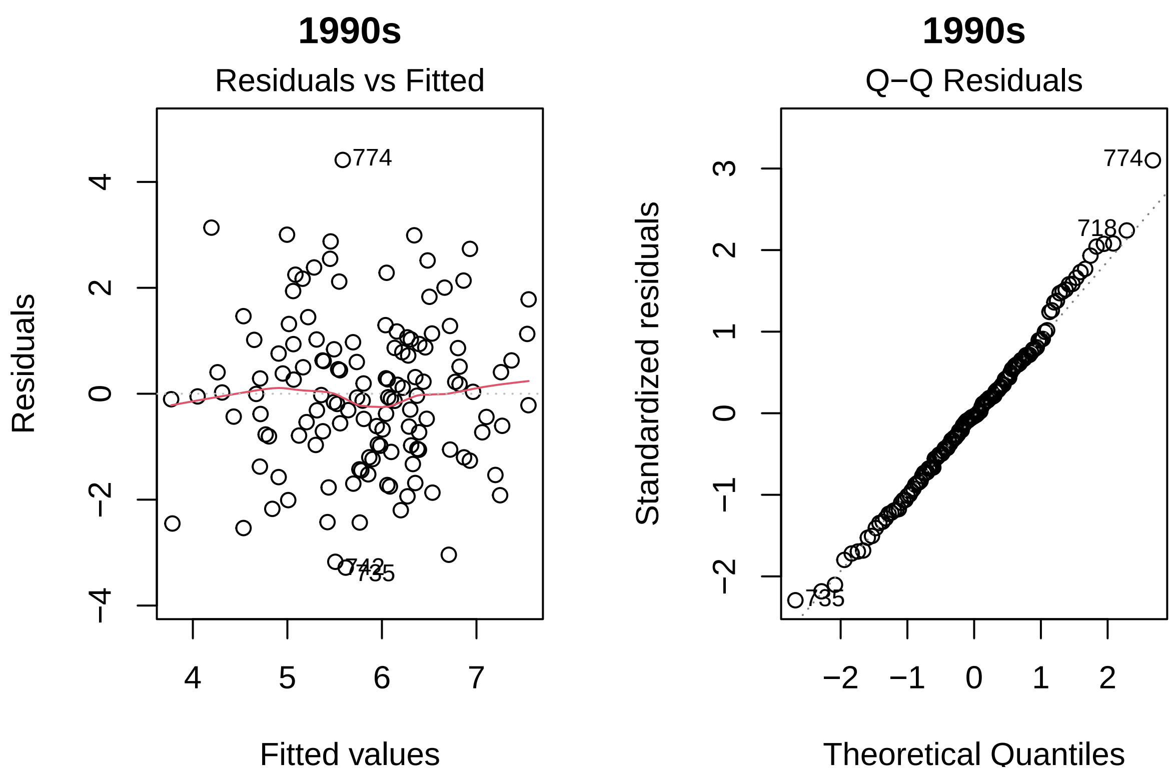
```
## Warning: not plotting observations with leverage one:
##   239
```




```
plot(model_80s, which = 1:2, main = "1980s")
```

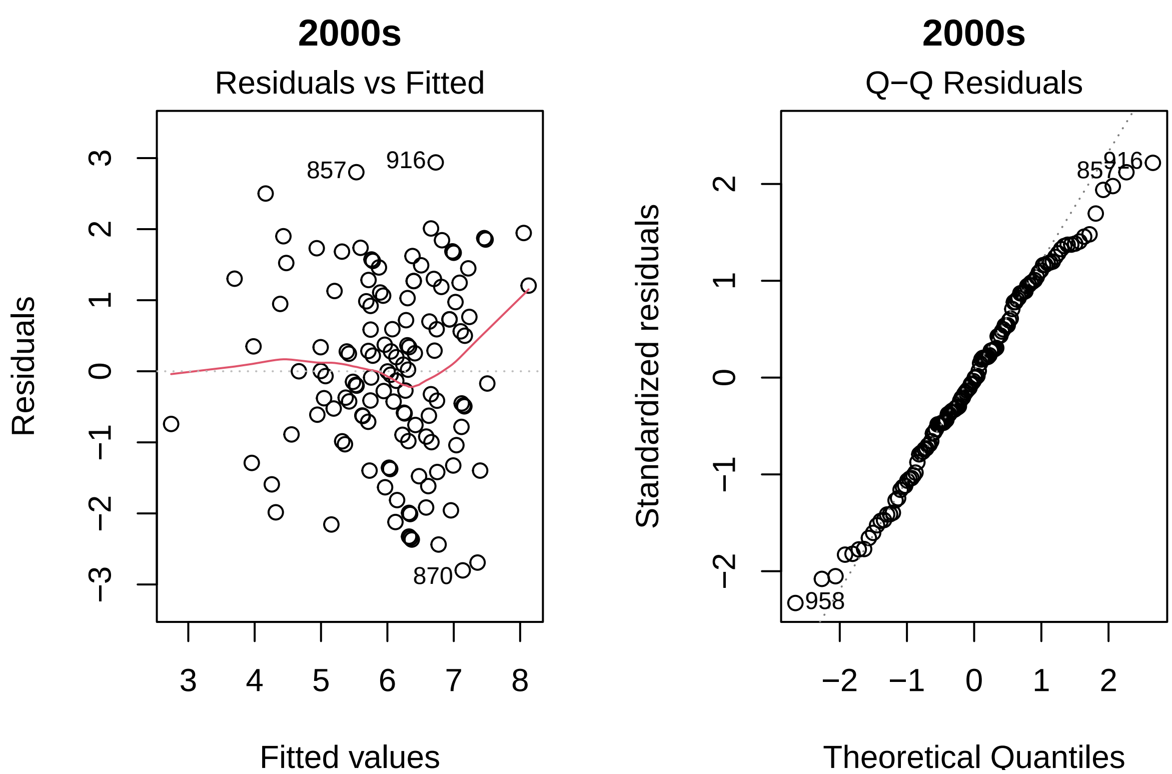



```
plot(model_90s, which = 1:2, main = "1990s")
```

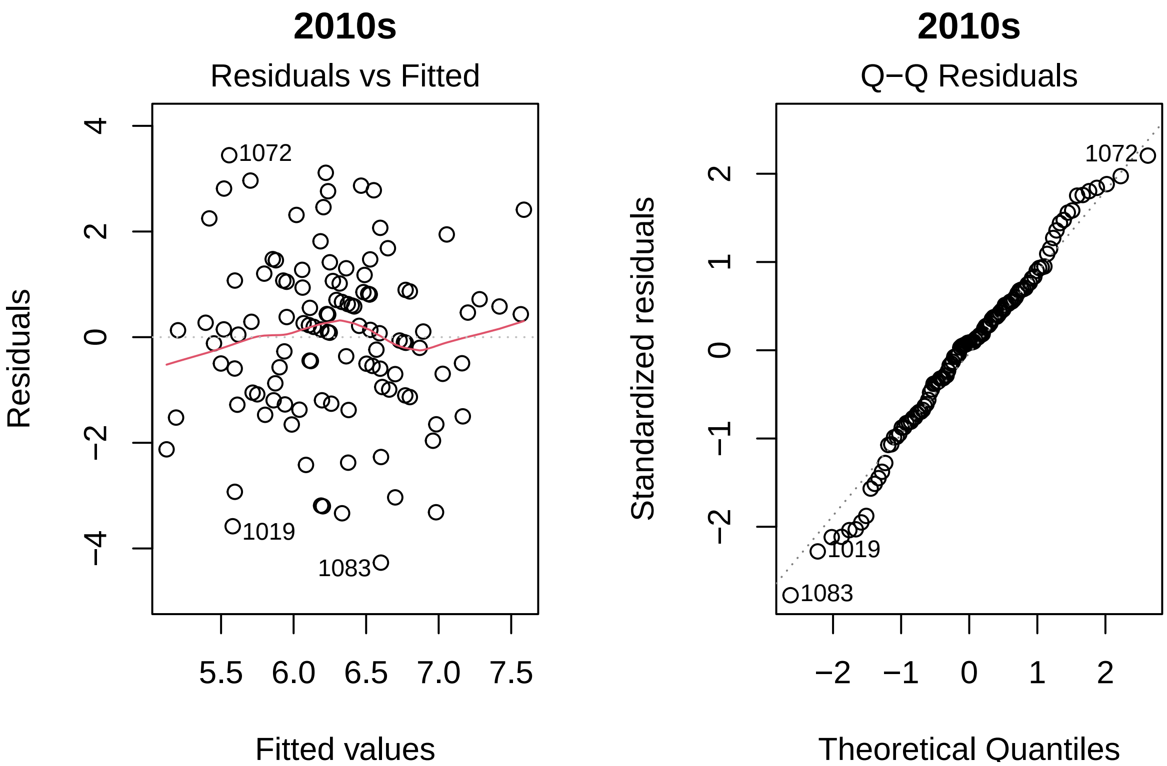



```r
plot(model_2000s, which = 1:2, main = "2000s")
```

```
## Warning: not plotting observations with leverage one:
##   4
```



```r
plot(model_2010s, which = 1:2, main = "2010s")
```



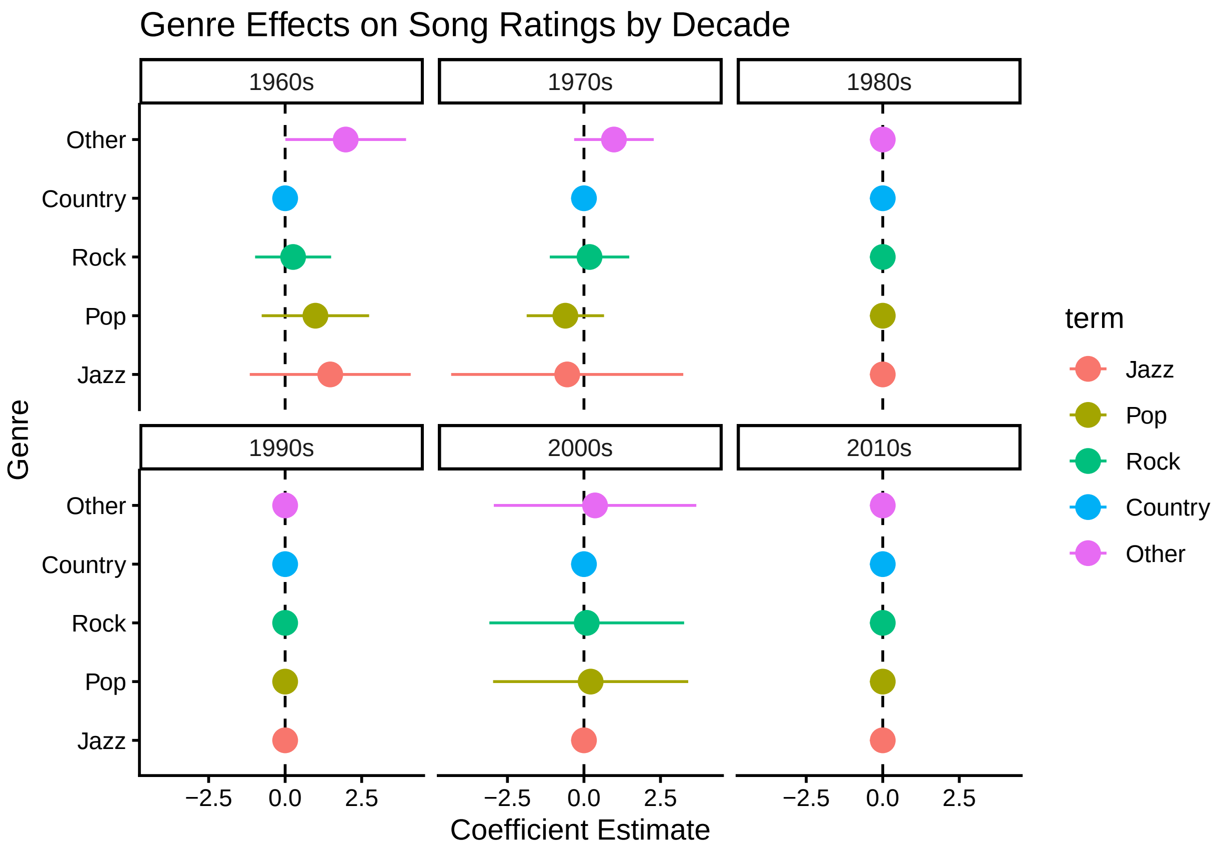
The fitted regression lines indicate that the relationship between song length and rating varies by genre. In particular, certain genres (e.g., Other and Country) exhibit steeper positive slopes, while others (e.g., Rock and Pop) show weaker or slightly negative trends, suggesting a potential interaction between length and genre.

```r
library(broom)
```

```
## Warning: package 'broom' was built under R version 4.4.3
```

```r
coef_1960s <- tidy(model_60s, conf.int = TRUE) %>% mutate(decade = "1960s")
coef_1970s <- tidy(model_70s, conf.int = TRUE) %>% mutate(decade = "1970s")
coef_1980s <- tidy(model_80s, conf.int = TRUE) %>% mutate(decade = "1980s")
coef_1990s <- tidy(model_90s, conf.int = TRUE) %>% mutate(decade = "1990s")
coef_2000s <- tidy(model_2000s, conf.int = TRUE) %>% mutate(decade = "2000s")
coef_2010s <- tidy(model_2010s, conf.int = TRUE) %>% mutate(decade = "2010s")


#Appending Data not Represented
#1960s
coef_1960s <- rbind(coef_1960s, list("genreCountry", 0, 0, 0, 0, 0, 0, "1960s"))

#1970s
coef_1970s <- rbind(coef_1970s, list("genreCountry", 0, 0, 0, 0, 0, 0, "1970s"))

#1980s
coef_1980s <- rbind(coef_1980s, list("genreCountry", 0, 0, 0, 0, 0, 0, "1980s"))
coef_1980s <- rbind(coef_1980s, list("genreOther", 0, 0, 0, 0, 0, 0, "1980s"))
coef_1980s <- rbind(coef_1980s, list("genreJazz", 0, 0, 0, 0, 0, 0, "1980s"))
coef_1980s <- rbind(coef_1980s, list("genreRock", 0, 0, 0, 0, 0, 0, "1980s"))
coef_1980s <- rbind(coef_1980s, list("genrePop", 0, 0, 0, 0, 0, 0, "1980s"))
```

```r
#1990s
coef_1990s <- rbind(coef_1990s, list("genreCountry", 0, 0, 0, 0, 0, 0, "1990s"))
coef_1990s <- rbind(coef_1990s, list("genreOther", 0, 0, 0, 0, 0, 0, "1990s"))
coef_1990s <- rbind(coef_1990s, list("genreJazz", 0, 0, 0, 0, 0, 0, "1990s"))
coef_1990s <- rbind(coef_1990s, list("genreRock", 0, 0, 0, 0, 0, 0, "1990s"))
coef_1990s <- rbind(coef_1990s, list("genrePop", 0, 0, 0, 0, 0, 0, "1990s"))


#2000s
coef_2000s <- rbind(coef_2000s, list("genreCountry", 0, 0, 0, 0, 0, 0, "2000s"))
coef_2000s <- rbind(coef_2000s, list("genreJazz", 0, 0, 0, 0, 0, 0, "2000s"))

#2010s
coef_2010s <- rbind(coef_2010s, list("genreCountry", 0, 0, 0, 0, 0, 0, "2010s"))
coef_2010s <- rbind(coef_2010s, list("genreOther", 0, 0, 0, 0, 0, 0, "2010s"))
coef_2010s <- rbind(coef_2010s, list("genreJazz", 0, 0, 0, 0, 0, 0, "2010s"))
coef_2010s <- rbind(coef_2010s, list("genreRock", 0, 0, 0, 0, 0, 0, "2010s"))
coef_2010s <- rbind(coef_2010s, list("genrePop", 0, 0, 0, 0, 0, 0, "2010s"))




coef_data <- bind_rows(
  coef_1960s,
  coef_1970s,
  coef_1980s,
  coef_1990s,
  coef_2000s,
  coef_2010s
)

coef_data <- coef_data %>%
  filter(term != "(Intercept)") %>%
  filter(grepl("^genre", term)) %>%
  mutate(term = gsub("^genre", "", term))

coef_data$term <- factor(
  coef_data$term,
  levels = c("Jazz","Pop","Rock","Country","HipHop","Other")
)

ggplot(coef_data, aes(x = estimate, y = term, color = term)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_pointrange(aes(xmin = conf.low, xmax = conf.high), size = 0.8) +
  facet_wrap(~ decade) +
  theme_classic() +
  labs(
    title = "Genre Effects on Song Ratings by Decade",
    x = "Coefficient Estimate",
    y = "Genre"
  )
```



Country will always be 0 since it is our reference level.

```
model_60s
```

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)
##
## Coefficients:
##                (Intercept)                          age
##                   6.235892                    -0.036813
##             genderNot Male                    raceWhite
##                  -1.820687                     1.604179
##                  genreJazz                   genreOther
##                   1.473039                     1.978211
##                   genrePop                    genreRock
##                   0.988531                     0.260014
##                     energy                 danceability
##                  -0.005765                    -0.013030
##                  happiness                   length_sec
##                   0.004845                     0.008965
##        raceWhite:genreJazz         raceWhite:genreOther
##                  -1.353373                    -3.249204
##         raceWhite:genrePop          raceWhite:genreRock
##                  -1.786329                           NA
```

```
##            raceWhite:energy       raceWhite:danceability
##                    0.024399                    -0.046982
## genderNot Male:danceability     genderNot Male:happiness
##                    0.076584                    -0.030694
```

```
model_70s
```

```
##
## Call:
## lm(formula = rating ~ age + race + genre + length_sec, data = subpop)
##
## Coefficients:
## (Intercept)          age    raceWhite    genreJazz   genreOther     genrePop
##    5.250133    -0.033100    -0.430183    -0.547224     0.979427    -0.607790
##   genreRock   length_sec
##    0.182210     0.009494
```

```
model_80s
```

```
##
## Call:
## lm(formula = rating ~ energy + danceability + happiness, data = subpop)
##
## Coefficients:
##  (Intercept)        energy  danceability     happiness
##     4.960976      0.009412      0.015115     -0.011560
```

```
model_90s
```

```
##
## Call:
## lm(formula = rating ~ race + gender + energy + happiness + loudness_db,
##     data = subpop)
##
## Coefficients:
##    (Intercept)       raceWhite  genderNot Male          energy       happiness
##        9.46577        -0.46920         0.72094        -0.03512         0.01694
##    loudness_db
##        0.31154
```

```
model_2000s
```

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + bpm + length_sec + age:danceability +
##     race:genre + race:danceability + race:happiness + race:bpm +
##     race:length_sec + gender:danceability + gender:happiness +
##     gender:bpm, data = subpop)
##
## Coefficients:
##                 (Intercept)                          age
##                   14.254799                    -0.262392
##              genderNot Male                    raceWhite
##                   -0.551642                   -13.479191
##                  genreOther                     genrePop
##                    0.362481                     0.218854
```

```
##                    genreRock                          energy
##                     0.090807                        0.018882
##                 danceability                       happiness
##                    -0.083852                        0.001354
##                          bpm                      length_sec
##                    -0.014549                       -0.009048
##             age:danceability            raceWhite:genreOther
##                     0.003737                       -1.186109
##           raceWhite:genrePop              raceWhite:genreRock
##                           NA                              NA
##       raceWhite:danceability             raceWhite:happiness
##                     0.084078                       -0.025028
##                raceWhite:bpm            raceWhite:length_sec
##                     0.035295                        0.019703
## genderNot Male:danceability     genderNot Male:happiness
##                    -0.041970                        0.021226
##           genderNot Male:bpm
##                     0.019019
```

```r
model_2010s
```

```
##
## Call:
## lm(formula = rating ~ gender + energy + happiness + length_sec,
##     data = subpop)
##
## Coefficients:
##    (Intercept)  genderNot Male          energy       happiness      length_sec
##       4.788777        0.459667       -0.024064        0.012193        0.009668
```

```r
anova(model_results[["1960s"]]$first, model_results[["1960s"]]$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##     danceability + age * happiness + age * bpm + age * loudness_db +
##     age * length_sec + race * genre + race * energy + race *
##     danceability + race * happiness + race * bpm + race * loudness_db +
##     race * length_sec + gender * genre + gender * energy + gender *
##     danceability + gender * happiness + gender * bpm + gender *
##     loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1    183 673.25
## 2    155 535.98 28    137.27 1.4178 0.09454 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(model_results[["1970s"]]$first, model_results[["1970s"]]$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
```

```
##     danceability + age * happiness + age * bpm + age * loudness_db +
##     age * length_sec + race * genre + race * energy + race *
##     danceability + race * happiness + race * bpm + race * loudness_db +
##     race * length_sec + gender * genre + gender * energy + gender *
##     danceability + gender * happiness + gender * bpm + gender *
##     loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    231 764.63
## 2    204 666.81 27    97.812 1.1083 0.3327
```

```
anova(model_results[["1980s"]]$first, model_results[["1980s"]]$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##     danceability + age * happiness + age * bpm + age * loudness_db +
##     age * length_sec + race * genre + race * energy + race *
##     danceability + race * happiness + race * bpm + race * loudness_db +
##     race * length_sec + gender * genre + gender * energy + gender *
##     danceability + gender * happiness + gender * bpm + gender *
##     loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    216 494.88
## 2    192 459.91 24    34.968 0.6083 0.9247
```

```
anova(model_results[["1990s"]]$first, model_results[["1990s"]]$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##     danceability + age * happiness + age * bpm + age * loudness_db +
##     age * length_sec + race * genre + race * energy + race *
##     danceability + race * happiness + race * bpm + race * loudness_db +
##     race * length_sec + gender * genre + gender * energy + gender *
##     danceability + gender * happiness + gender * bpm + gender *
##     loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    122 269.95
## 2     98 217.50 24    52.448 0.9846  0.493
```

```
anova(model_results[["2000s"]]$first, model_2000s)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + danceability +
##     happiness + bpm + length_sec + age:danceability + race:genre +
##     race:danceability + race:happiness + race:bpm + race:length_sec +
##     gender:danceability + gender:happiness + gender:bpm
##   Res.Df    RSS Df Sum of Sq     F    Pr(>F)
## 1    116 270.33
```

```
## 2    108 208.61  8    61.719 3.994 0.0003515 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(model_2000s, model_results[["2000s"]]$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + danceability +
##     happiness + bpm + length_sec + age:danceability + race:genre +
##     race:danceability + race:happiness + race:bpm + race:length_sec +
##     gender:danceability + gender:happiness + gender:bpm
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##     danceability + age * happiness + age * bpm + age * loudness_db +
##     age * length_sec + race * genre + race * energy + race *
##     danceability + race * happiness + race * bpm + race * loudness_db +
##     race * length_sec + gender * genre + gender * energy + gender *
##     danceability + gender * happiness + gender * bpm + gender *
##     loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    108 208.61
## 2     93 194.20 15    14.415 0.4602 0.9544
```

```r
anova(model_results[["2010s"]]$first, model_2010s)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##     happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ gender + energy + happiness + length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    103 274.94
## 2    110 279.47 -7   -4.5351 0.2427 0.9735
```

```r
anova(model_2010s, model_results[["2010s"]]$second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ gender + energy + happiness + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##     danceability + age * happiness + age * bpm + age * loudness_db +
##     age * length_sec + race * genre + race * energy + race *
##     danceability + race * happiness + race * bpm + race * loudness_db +
##     race * length_sec + gender * genre + gender * energy + gender *
##     danceability + gender * happiness + gender * bpm + gender *
##     loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    110 279.47
## 2     81 206.34 29    73.132 0.9899 0.4938
```